Matt Romano

215-378-6942 | mmr6806@psu.edu | linkedin.com/in/matthew-romano913/ | github.com/MatthewDRomano

Education

The Pennsylvania State University – University Park State College, PA

Bachelor of Science in Computer Science | Cumulative GPA: 3.97 / 4.00 (Junior) *Expected Grad. May 2028*

- **Minors:** Computer Engineering, Cybersecurity, Mathematics
- **Relevant Coursework:** Computer Architecture, Computer Organization & Design (MIPS Assembly, Verilog), Systems Programming, Computer Communication Networks, Data Structures & Algorithms
- **Activities:** Member, IEEE Penn State student chapter

Technical Skills

Languages: C, Python, Verilog, MIPS Assembly, Bash, TCL, Java, SQL, C#, Scheme

Systems & Networking: Linux systems programming, POSIX threading, C11 atomics, Socket API, inotify, uthash

Hardware & FPGA: Pipelined datapath design, hazard/forwarding units, RTL verification, FPGA synthesis

Embedded & Low-Level: Bare-metal firmware, register-level peripheral bring-up, SPI, memory-mapped I/O, GRUB

Tools & Environments: Vivado, Zynq-7000 SoC, GDB, Valgrind, Git, make, Docker, Vim, VS Code, Visual Studio

Experience / Projects

dirsyncd | *C, POSIX/Linux (inotify), uthash* July 2026 – Aug. 2026

- Built a Linux directory-sync daemon in C that mirrors file changes between directories as they happen, using filesystem events instead of repeatedly scanning the directory tree.
- Replaced timer-based full-tree diffing with `inotify` triggers, cutting idle CPU cost to zero, with a uthash registry giving O(1) lookup up to the kernel's watch limit.
- Stress-tested at thousands of directory creates and deletes per second with no missed changes, rebuilding every watch if the kernel drops events so the daemon never acts on a partial view.

AtomiC-IO Broadcast Server & Client API | *C, POSIX Threads / Sockets* Jan. 2026 – July 2026

- Built a C library that lets programs exchange messages over TCP, including broadcasting a single message from a server to every connected client.
- Sustained 128 concurrent clients across multiple machines under flood-rate messaging with zero lost messages, using a `poll()`-based server with background consumer threads.
- Root-caused an intermittent hang and segfault under GDB and Valgrind to a byte-order bug overflowing a receive-thread stack buffer by roughly 1,020 bytes, compounded by a discarded `atomic_exchange` return.
- Coordinated shared connection state with C11 compare-and-swap and fine-grained mutexes, and validated all 4 modules with a dependency-free test harness.

5-Stage Pipelined MIPS Processor | *Verilog, Vivado, Zynq-7000 FPGA* Jan. 2026 – Aug. 2026

- Designed a working CPU from scratch in Verilog that runs MIPS programs, supporting 30+ instructions and overlapping up to 5 at once in a pipeline, synthesized to a Zynq-7000 FPGA.
- Built the forwarding, stalling, and branch-resolution units that keep the pipeline correct when instructions depend on each other, resolving branches in EXE so a taken branch costs a 2-instruction flush.
- Wrote automated Verilog test benches checking fetch, ALU, memory, and write-back against expected state, tracing cascading X/z failures in simulation to floating signals in the harness.

nRF52840 Sensor Firmware | *C, Nordic nRF52840, LSM6DSO* Aug. 2026 – Present

- Building bare-metal firmware for a Nordic microcontroller that reads motion data from an LSM6DSO accelerometer and gyroscope over SPI, without vendor libraries or example code.
- Writing SPIM peripheral initialization directly from the nRF52840 register-level datasheet and the LSM6DSO register map, debugging the hardware and firmware boundary directly.

Other Experience

Tennis/Pickleball Court Maintenance May 2024 – Aug. 2024

Northampton Tennis and Fitness Club *Richboro, PA*

- Co-led a 5-person task team preparing professional court-tracking bounds for corporate events hosting 50+ attendees.
- Reduced turnaround time on complex tennis-to-pickleball court conversions under tight scheduling constraints by working directly with management on layout optimization.